An research on the load sharing problem of two synchronous generators -

>>> This research contain that how the two generator which has different power generating capacity can share the load. Here we have different points on which we are going to discuss such as -

- Mechanical power input
- Governing of rotor
- Role of internal resistance
- Role of current division
- Frequency drop due to load
- Droop Characteristics
- Role of load angle in power sharing

Note : Because it is highly relevant to our above topic that is why -

Then after it we will also discuss little bit about that why we need to use the grid forming inverters or the ***Virtual Synchronous Generator (VSG)***. Here we have several topics to discuss such as - Why we need VSG, Constant frequency Problem, sudden load on the normal gird flowing generator and energy storage system to simulate the energy trade of like speed of the rotor reduce.

So lets start with our question -

/// Suppose there is two synchronous generator connected in parallel which has same frequency, same phase and same voltage. One has the rating of 10 megawatt and the other has the rating of 100 megawatt. We connected a load to the busbar from which the two generators are connected. So the load is around 30 megawatt. My question is that how they share the load, the generator share the load. Technically both have the same voltage, same frequency, so they should share the same load, for example 15 megawatt each. But as we know that generator 1 has the capacity of around 10 megawatt, then how the generator 1 limiting itself to generate only 10 megawatt?

>>>Another Followup question

Suppose two synchronous generators with different power ratings (10 MW and 100 MW) are connected in parallel to a common busbar. Both generators are properly synchronized, operating at the same voltage and frequency.

The total installed generation capacity is 110 MW. Therefore, Generator 1 represents approximately

9.09% of the total capacity, while Generator 2 represents approximately 90.91%.

If a load of 30 MW is connected to the common busbar, shouldn't the generators automatically share the load in proportion to their respective rated capacities?

In other words, should Generator 1 supply approximately 9.09% of the total load (2.73 MW), while Generator 2 supplies the remaining 90.91% (27.27 MW)?

If so, what mechanism ensures this proportional load sharing in practice?

/// The answer for this above question is that when two synchronous generator which has different power rating the load sharing is controlled by the steam governor.

>>>Suppose we want to impose less load on one generator and more load on the other generator then we simply reduce the steam output or little bit close the steam valve of generator one which reduce the mechanical input on the rotor which create the reduce in speed or drop in the frequency now there is

the important part the reduce in the frequency will reduce the bus-bar frequency if the bus-bar frequency reduce the other generator frequency will also reduce so the other generator increase its mechanical input on the rotor or increase the steam pressure which it means our other generator start delivering more power and maintain the bus-bar frequency.

>>> The relationship between the frequency drop and the power output of due to that frequency drop is calculated by the droop frequency.

Frequency droop is a control mechanism that allows multiple generators to automatically share an increasing or decreasing load without continuously communicating with each other.

Suppose we want to impose less load on one generator and more load on the other generator then we simply reduce the steam output or little bit close

the steam valve of generator one which reduce the mechanical input on the rotor which create the

A governor is designed so that the generator's power output changes according to the frequency.

Frequency	Generator output
50.0 Hz	0 MW
49.75 Hz	25 MW
49.5 Hz	50 MW
49.25 Hz	75 MW
49.0 Hz	100 MW

The mathematical expression is:

$$R = \frac{\Delta f / f_n}{\Delta P / P_n}$$

Where:

- R = Droop percentage (in per unit)
- Δf = Change in frequency
- f_n = Rated frequency
- ΔP = Change in power
- P_n = Rated power

Suppose we have two synchronous generators connected in parallel.

Parameter	Generator 1	Generator 2
Rated capacity	100 MW	50 MW
Droop	5%	5%
Rated frequency	50 Hz	50 Hz

Both generators are operating together.

Their droop characteristics are:

For Generator 1:

$$5\% \times 50 = 2.5 \text{ Hz}$$

Therefore, a frequency reduction of 2.5 Hz corresponds to a 100 MW power increase.

For Generator 2:

$$5\% \times 50 = 2.5 \text{ Hz}$$

A frequency reduction of 2.5 Hz corresponds to a 50 MW power increase.

Their power-frequency relationships are:

$$P_1 = \frac{100}{2.5}(50 - f)$$

$$\boxed{P_1 = 40(50 - f)}$$

Similarly:

$$P_2 = \frac{50}{2.5}(50 - f)$$

$$\boxed{P_2 = 20(50 - f)}$$

Here, power is in MW and frequency is in Hz.

Now suppose the load increases.

Let's say the total load is 30 MW.

For simplicity, assume both generators initially have zero output at 50 Hz.

When the load increases, frequency starts decreasing.

Suppose the system reaches 49.5 Hz.

Generator 1:

$$P_1 = 40(50 - 49.5)$$

$$\boxed{P_1 = 20 \text{ MW}}$$

Generator 2:

$$P_2 = 20(50 - 49.5)$$

$$\boxed{P_2 = 10 \text{ MW}}$$

Total generation:

$$P_{total} = 20 + 10$$

$$\boxed{P_{total} = 30 \text{ MW}}$$

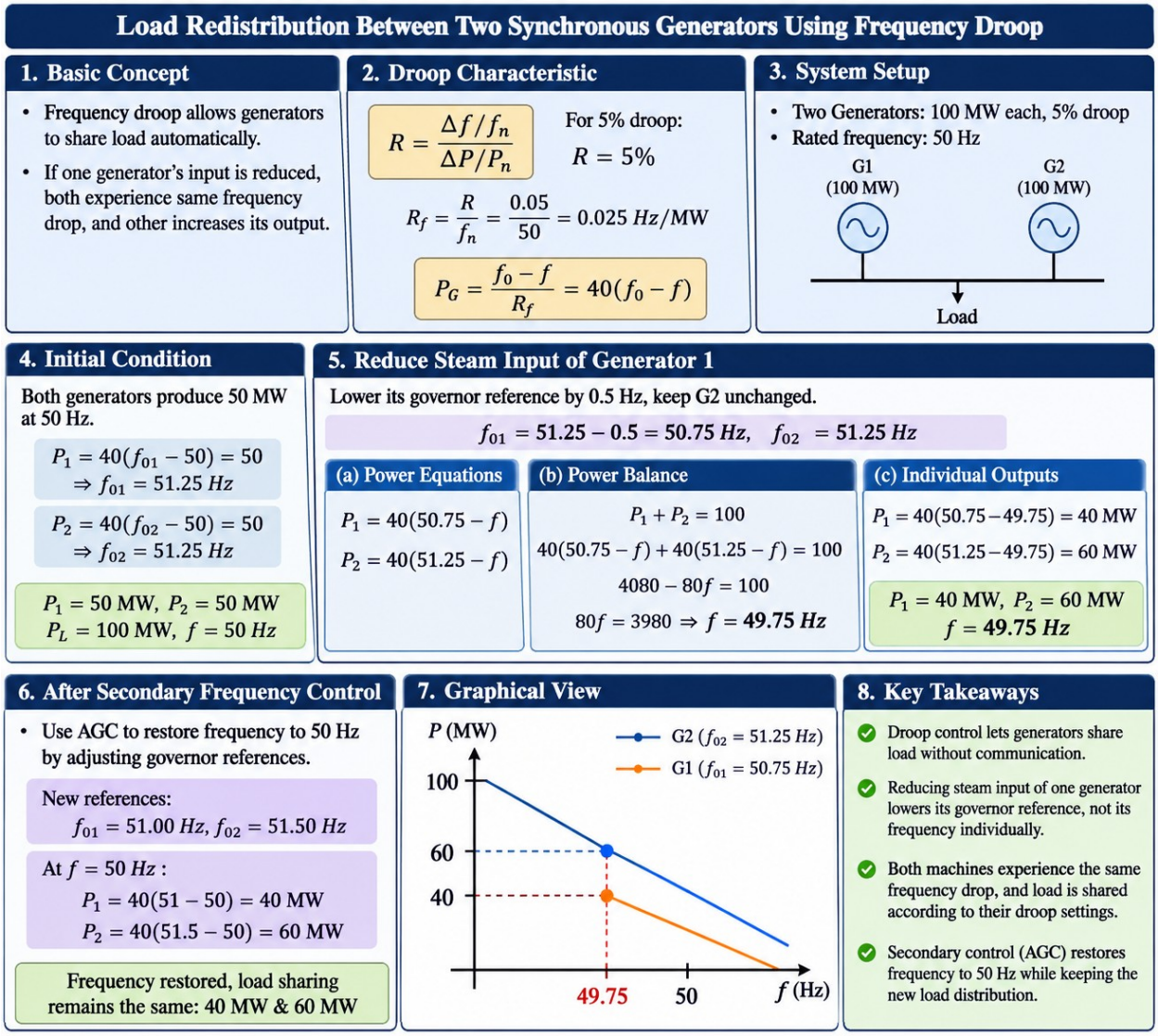
Look carefully!

- Generator 1 supplies 20 MW.
- Generator 2 supplies 10 MW.
- Total load is 30 MW.

/// Here you can see different generator has different Droop characteristic according to there load.

//##// The droop characteristics of a synchronous generator are primarily determined by the settings and design of its governor control system rather than by its manufacturing date or the connected electrical load. The governor controls the mechanical input supplied to the prime mover (such as steam, water, or fuel) according to changes in system frequency. Engineers configure the governor's droop setting during commissioning or operation to achieve the desired load-sharing behaviour. Different generators can have different droop settings depending on their rated capacities, governor designs, prime mover characteristics, and grid operating requirements. Consequently, when multiple generators are connected to the same grid, they experience the same frequency deviation but adjust their mechanical input and active power output according to their individual droop characteristics. Thus, the connected load determines the

total power demand, whereas the governor droop settings and generator ratings determine how that demand is shared among the generators.



Now lets talk about the power angel that what is relation of the power angle with the governing and the load sharing .

>> The power angle equations tell us that represent the power transfer from the generator to

the grid where the anger which is delta represent the physical angel between the grid voltage and the generated voltage.

>> In other words to transfer the electricity form the generator to the grid the power angle should we increase here the increase in power angle means the rotor should rotate little bit ahead in compare with the grid voltage to transfer the power.

>> The key thing to remember here is that the frequency and the speed remain same relatively but the angel between the gird voltage and the rotor magnetic reference is different and the angular position of the determines how much active electrical power generator transfer.

These three equations describe different parts of the same physical process.

Equation	What does it explain?
$P = \frac{EV}{X_s} \sin \delta$	How the power angle determines electrical power transfer.
$P = T\omega$	Relationship between mechanical torque and mechanical power.
$J\frac{d\omega}{dt} = T_m - T_e$	How the rotor responds when mechanical and electromagnetic torques are unequal.

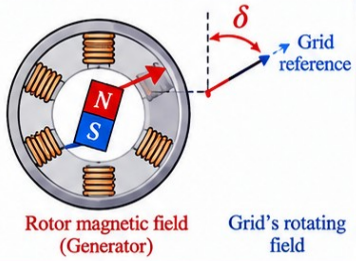
Stage	Frequency droop	Swing equation	Power-angle equation
Reduce steam input to G1	G1 power reference decreases	G1 initially decelerates	G1 angle decreases
System frequency decreases	G2 governor detects the decrease	G2 accelerates relative to the bus	G2 angle increases
Load redistribution	G1 output decreases, G2 output increases	Both reach a new equilibrium	G1 has a smaller angle, G2 a larger angle
Final condition	Governors maintain new power sharing	No continuing acceleration	Both angles remain constant

Power-Angle Characteristics and Load Sharing in Synchronous Generators

Understanding the Physical Meaning – From Rotor Angle to Power Transfer

1. What is Power Angle (δ)?

It is the angular displacement between the generator's internal magnetic field and the grid's rotating reference.



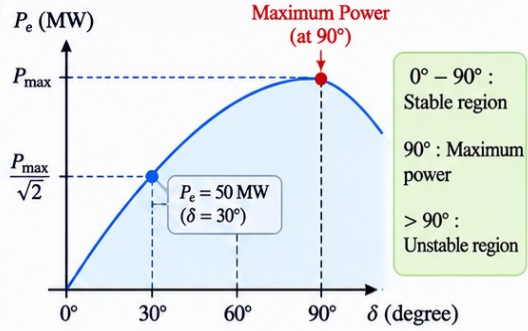
2. Power-Angle Equation

$$P_e = \frac{EV}{X}\sin\delta = P_{max}\sin\delta$$

It shows how much electrical power the generator can deliver for a given power angle.

Key Point:
Increasing $\delta \rightarrow$ Increases P_e
(if $\delta < 90°$)

3. Power-Angle Curve



4. How δ Changes Physically?

(a) Increase Steam Input

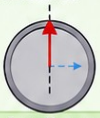
$P_m > P_e$
→ Rotor accelerates
→ δ increases



More Power (Larger δ)

(b) Decrease Steam Input

$P_m < P_e$
→ Rotor decelerates
→ δ decreases



Less Power (Smaller δ)

(c) Balance

$P_m = P_e$
→ δ is constant



Steady Operating Point

5. Two Generators in Parallel – Load Sharing

Both generators see the same frequency change, and adjust their power via δ.

Initially: 50 MW each	After Reducing G1's Steam
$\delta_1 = \delta_2 = 30°$	$\delta_1 = 23.58°$, $P_1 = 40$ MW
$P_1 = P_2 = 100\sin 30° = 50$ MW	$\delta_2 = 36.87°$, $P_2 = 60$ MW

G1's δ decreases → less power,
G2's δ increases → more power

6. Swing Equation Connection

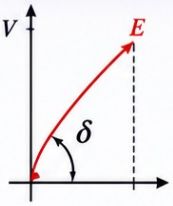
$$M\frac{d^2\delta}{dt^2} = P_m - P_e$$

- If $P_m > P_e \rightarrow \delta$ increases (rotor advances)
- If $P_m < P_e \rightarrow \delta$ decreases (rotor falls behind)

↓

Change in mechanical input leads to change in power angle, which changes electrical power.

7. Why More δ Gives More Power?



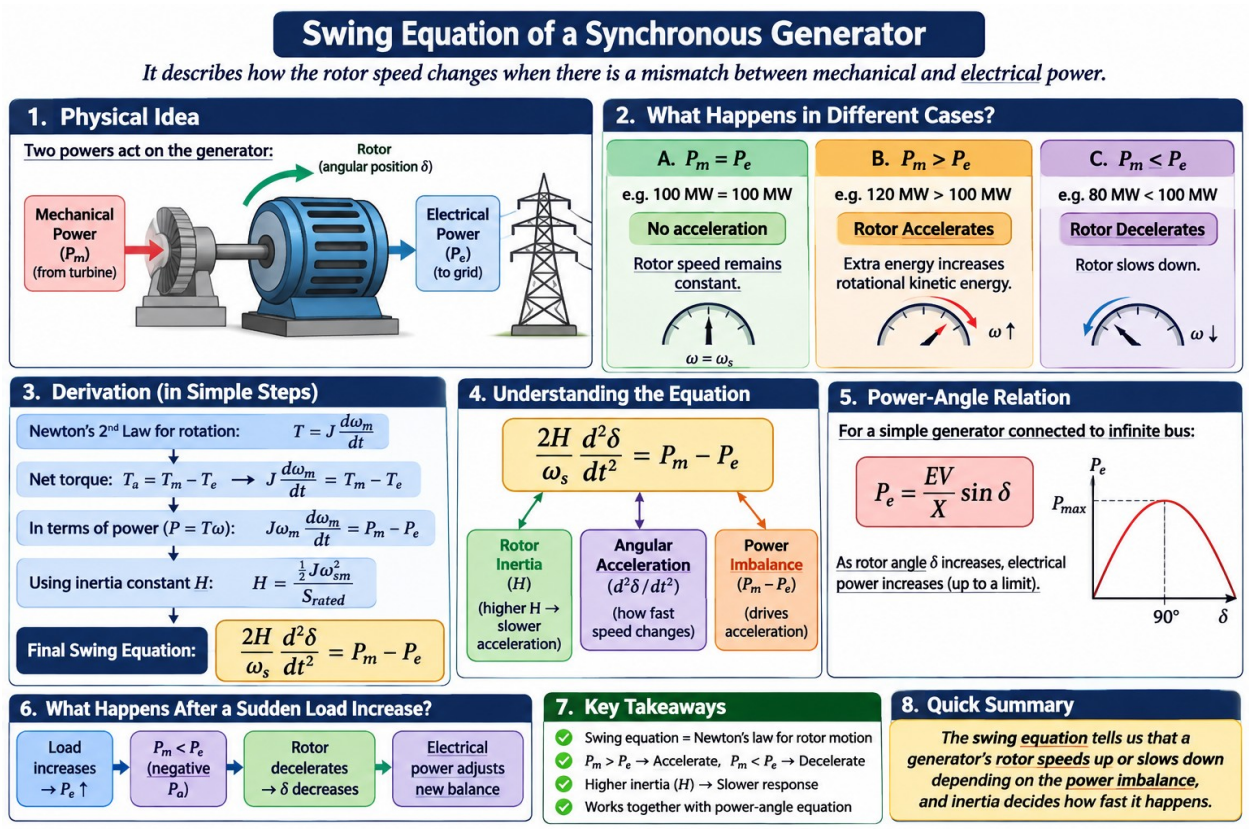
$$P \propto \sin\delta$$

A larger angle means a greater voltage component in the direction of the grid, so more power flows to the grid.

8. Key Takeaways

- Power angle (δ) is the rotor's angular position relative to the grid.
- $P_e = P_{max}\sin\delta$ shows how δ controls power transfer.
- Increasing mechanical input → δ increases → more power.
- Decreasing mechanical input → δ decreases → less power.
- In parallel generators, droop control adjusts mechanical input so that each generator gets its share of load, while operating at the same frequency.
- Power-angle characteristic is the bridge between governor control and electrical power transfer.

/// The swing equation the power system can also related to this concept because the swing equation determine the change in physical power of the synchronous generator like rotor inertia and the angular acceleration due to the variation in the electrical load .



Now lets understand that another most important concept which is related to the topics that we discussed above and has very important significance in our new power system.

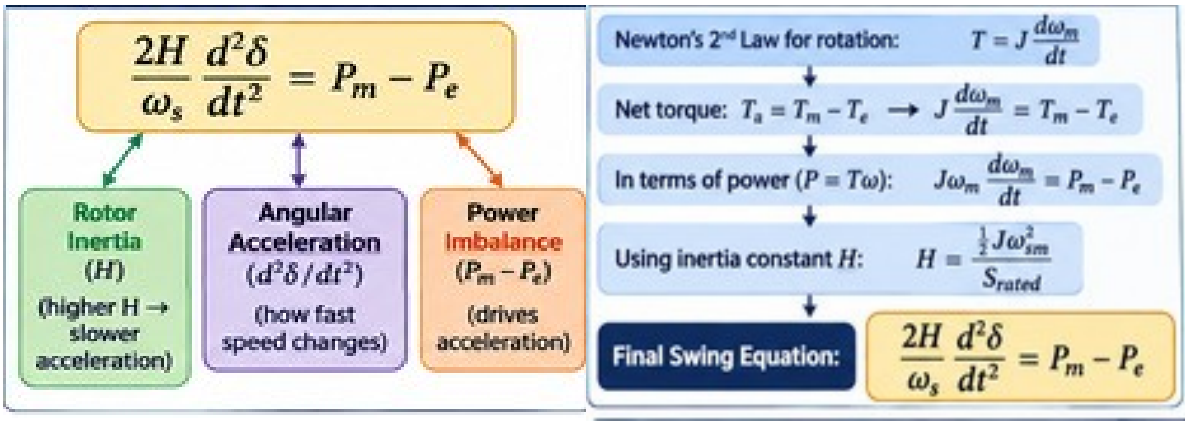
Virtual Synchronous Generators -

>>> This is not any physical machine instead these are the computer program or simulations which simulate the our traditional grid generator but virtually to stabilize the grid specially the solar dominant the gird.

/// Tradition Physical Grid which has rotating intertia in synchronous generator :

>>> We all know that when there is sudden load come on the grid the synchronous generators rotor speed start reducing which is detected by the sensors and the the steam flow will increase to maintain the speed and frequency here the rotor speed is traded off with the extra energy needed to the sudden load.

>>> In other word the rotor speed reduced provide the extra sudden energy and prevent the voltage from dropping and provide the stability to the grid which we can determine by the swing equation.



/// Grid which do not have the physical rotation inertial :

>>> Suppose we have an grid which only have the solar generating station but there is one gird forming inverter *(Remember not the virtual synchronous machine) which generate an constant frequency of 50Hz and other inverter flow it by the use of PLL (Method) method. This system sounds like an proper system because if there is not any problem of frequency drop when the sudden load impose on the grid the solar provide that power to the load if possible otherwise the load is disconnected or reduces from the grid that we generally done in the case of the overloaded grid.

But there is some very serious situation or problem

that we can face under this methodology which we discussed below -

1. The Load Sharing:

>>> As we already discussed that in the case two synchronous generators connected in parallel which has different power rating can share the load by the use of the drop characteristics and the frequency change due to the load.

>>>Now consider the same situation in the case of solar suppose we have the two solar station which has different output rating connected parallel now suppose there is an sudden load impose on the grid because both the generator working on the same voltage and same frequency then who will decide that how much energy each has to give.

**Remember i am considering that both the system do not have any mode of communication or has any centralized controlling system .*

In the case of synchronous generators we identify the frequency drop and throttle the turbine according to it but in the case of the solar we do not have any proper self governing or detection system, Here we

need to relay on the computer based software and controlled .

2. Sudden Load Distribution -

>>>When a large load is suddenly connected, the grid requires additional active If the grid-forming inverter maintains a fixed 50 Hz reference without an appropriate power-balancing mechanism, how will other inverters recognize the additional demand and respond?

3. Solar Generation Fluctuations -

>>>Solar generation continuously changes because of cloud movement, temperature and irradiance variations.
If one solar station suddenly loses generation, how will other stations automatically compensate without communication or frequency-based feedback?

4. Power Sharing During Generator Failure -

>>>Suppose the main grid-forming inverter suddenly disconnects because of a fault.

How will the remaining grid-following inverters maintain the grid voltage and frequency?
A conventional grid-following inverter depends on an existing voltage reference and cannot necessarily establish the grid independently.

Now lets discuss that how does the Virtual Synchronous Generators can solve this problem -

>>>The Virtual Synchronous Generators is nothing but an computer program which is simulating the synchronous generator properties and generate the signal to the inverters so that the inverters can behave like an synchronous generator such as -
Decrease the frequency according if the load increase, change the voltage, or even change the speed of the rotor of the virtual rotor that he simulated or vary the parameters according to load similar like the synchronous generator does.

/// Main advantages of the Virtual Synchronous generator -

>>> There major advantage of virtual synchronous generator is the load sharing and the system stability.

As we already discussed about that the

synchronous generators of two different rating share the load according to its droop characteristics which is entirely deppend on the synchronous generator design or power rating and the grid bus-bar frequency.

So suppose if we simulate the or convert an solar power plant into an virtual synchronous generator in which has its own virtual droop characteristics which work same like the original synchronous generator then we can solve the load sharing problem

Solar plant ---> Virtual Synchronous generator

Now understand that how does the virtual synchronous generator create the virtual rotor-

>>>When we look this question in more carefully then we find that it is very important question when we talk about the virtual synchronous generator because the rotor is the main part which provide the stability and the inertia to the gird so if we simulate the rotor then we can easily simulate the how generator

>>> Now the rotor of the actual synchronous

generator has the kinetic energy which he use to damp the sudden load or provide the extra energy under the sudden load.
for example if the load increase then the rotor speed reduce and provide the extra energy to the gird and maintain the voltage.

>>>But the computer simulation do not generate the extra energy by itself so to solve this problem we uses the most important part of the virtual synchronous generator which is an battery energy storage system. The virtual energy storage system provide that extra energy for the grid while the frequency decrease.

